

CROSSCURRENT

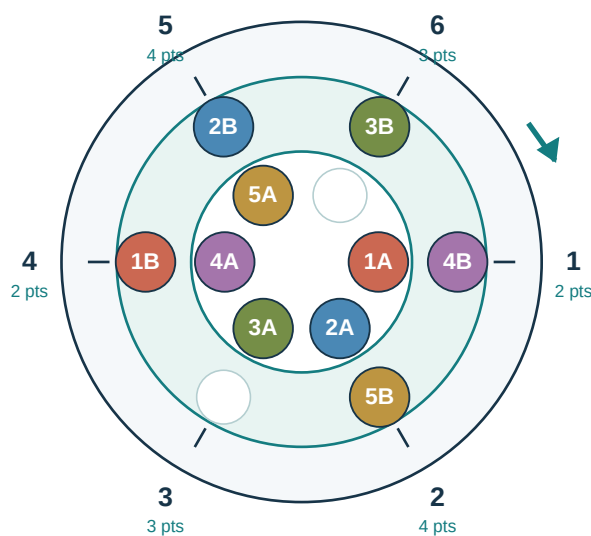
A shared-current harbor game for 2-5 players. Designed for ages 8+ and a 20-40 minute family session; age suitability, teaching time and duration are estimates awaiting human testing. Rules revision 1.

Choose a boat and an arrow in secret. Your arrows combine to turn whole rings of boats. Harvest at the shore, share crowded rewards, then cross into the other current. Score the most points after 12 rounds; tied players share the win.

In the box

- One fixed shore, one inner disk and one outer ring.
- Ten boat tokens: two for each of five players. A player's pair has the same engraved tally count, from one to five. Label one A and the other B. Color is optional; tally identity and letters identify every boat without it.
- Twenty-five paper cards: each player's set contains A, B, clockwise +1, still 0 and counterclockwise -1.
- A paper score sheet and pencil. For 2-4 players, put unused identities and their cards away.

The shore has six numbered harbors. Starting at harbor 1 and going clockwise, their rewards are **2, 4, 3, 2, 4, 3** points. Raised pips show rewards; the written numeral shows harbor identity. Both rings have one berth aligned to each harbor. The shore stays still throughout the game.



Five-player starting position

1A means player 1, boat A.

A boats begin on the inner disk.

B boats begin on the outer ring.

Harbor 1 is shown at the right.

2, 3, 4, 5 and 6 follow clockwise.

Number = harbor; points = reward.

Use the table for 2, 3 or 4 players.

Illustrated five-player setup. Rotate this entire view to match your seat; the harbor order remains clockwise.

This diagram may be rotated on the table; the clockwise order is what matters.

Set up

1. Place the shore flat. Drop the inner disk and outer ring into their circular guides, pocket sides up. Turn them until all twelve berth centers line up with the six shore marks. There is no special ring starting orientation: its berths are identical.
2. Choose player identities and assign setup seats from left to right in the table below. You may randomize who takes each seat. These seats set initial positions only; there is no first player.
3. Place each A on the inner disk and B on the outer ring at the listed harbor. Give each player their five cards. All three direction cards are available.
4. Write names and tally identities on the score sheet, set scores to zero and start round 1. Put all boat locations, available and used directions, and scores where everyone can inspect them.

Players	Seat 1: A / B	Seat 2: A / B	Seat 3: A / B	Seat 4: A / B	Seat 5: A / B
2	1 / 4	4 / 1	-	-	-
3	1 / 4	3 / 6	5 / 2	-	-
4	1 / 4	2 / 5	4 / 1	5 / 2	-
5	1 / 4	2 / 5	3 / 6	4 / 1	5 / 2

Every A in this table starts inner; every B starts outer. On later games, exchange setup seats.

A round: choose, turn, score, cross

1. Choose together

Secretly choose your **A or B** selector and **one available direction**. Put the two cards face down. Everyone locks in their pair before anyone reveals. You may inspect your own pair while choosing. Reveal all pairs together once everyone is ready.

Each three-round group is a **tide**. Use each direction exactly once in that tide. A used direction stays face up and cannot be chosen again until the tide ends. Your A and B selectors are always returned after the round; you may select the same boat repeatedly.

Talking and bluffing are allowed, but promises are not binding. Do not trade cards, boats or points. No player may see another player's current hidden pair.

2. Turn both currents

Before moving either ring, note which ring each selected boat occupies. That is the ring receiving that player's direction this round.

Add the revealed directions separately for the inner and outer rings. A +1 and -1 cancel; 0 adds nothing. Turn each ring by its total number of harbor steps, clockwise for a positive total and counterclockwise for a negative total, always viewed from above. A ring receiving no arrows stays still. **All boats aboard a ring travel with it, including boats that were not selected.**

For easy handling, use the equivalent turn of at most three steps:

Total	Turn
-5	1 clockwise
-4	2 clockwise
-3	3 either way
-2	2 counterclockwise
-1	1 counterclockwise
0	No turn
+1	1 clockwise
+2	2 clockwise
+3	3 either way
+4	2 counterclockwise
+5	1 counterclockwise

One person turns the rings while another checks the totals. Bring every berth back into line with a shore mark before scoring. Ring order makes no difference: arrows were assigned from the positions at reveal. A still card does not freeze a ring against other players' arrows.

3. Score the shore

Only the selected boat of each player harvests. At each harbor, count **all selected boats across both rings**. Divide that harbor's reward by the count and round down. Each selected boat there earns that number of points. Discard any remainder. Unselected boats neither score nor reduce anyone's reward.

Harbor reward	1 selected boat	2 selected boats	3 selected boats	4 selected boats	5 selected boats
2	2	1	0	0	0
3	3	1	1	0	0
4	4	2	1	1	0

Write every player's earnings before moving any boat across rings. Scores are public and never buy extra powers.

4. Cross and drift

Each selected boat now moves to the **other ring**, at the harbor **one step clockwise from the harbor where it just scored**. Harbor 6 wraps to harbor 1. Every crossing uses that same scoring snapshot. A boat moves once, even if another boat arrives beside it; arriving boats never score again this round. Unselected boats stay where their ring left them.

Occupied destinations never block a boat. Boats belonging to any players may share a berth. Stack them cup side up and inspect a stack freely whenever needed; return every inspected boat to its original ring and harbor. Stack order has no rules effect. When lifting a selected boat from beneath another, replace the unselected boats at the same berth.

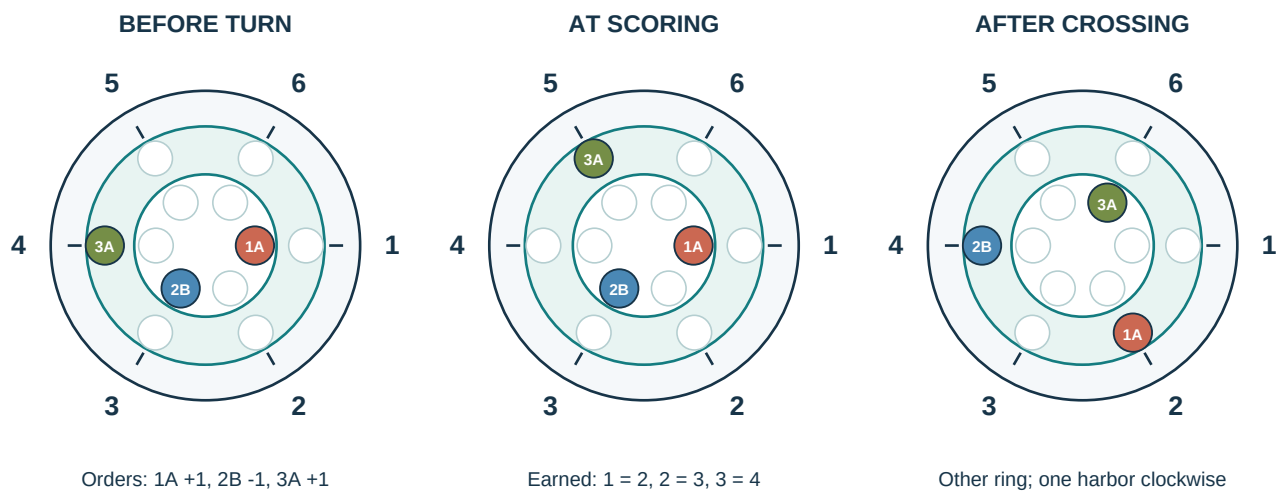
Return selector cards to hand and place the used directions face up. Advance the round. After rounds **3, 6 and 9**, return all three directions to every player's hand. After round **12**, the highest total wins; share a tie. There is no extra round or tiebreaker.

Worked round

This is an example position during a game, not the starting layout. Player 1 has one tally, Player 2 two and Player 3 three; colors may help, but are unnecessary.

Player	Before reveal	Arrow	Ring total	At scoring	Points	After crossing
1	A, inner harbor 1	+1	Inner: 0	Inner harbor 1	2	Outer harbor 2
2	B, inner harbor 3	-1	Inner: 0	Inner harbor 3	3	Outer harbor 4
3	A, outer harbor 4	+1	Outer: +1	Outer harbor 5	4	Inner harbor 6

The inner arrows cancel. The outer ring moves one step clockwise, carrying **every** outer boat, including unselected ones. The three selected boats score at different harbors, so no reward is shared. Then they cross and drift as shown. Used arrows stay unavailable until the next tide.



Only the three selected boats are shown here. Unselected boats still ride their rings; they do not harvest or cross.

For a crowd example, suppose selected boats end at inner harbor 2 and outer harbor 2. Harbor 2 pays four; they each earn two, even though they occupy different rings. A third **unselected** boat there changes nothing. If a third selected boat arrives there instead, each earns one. All selected boats then cross to the other ring at harbor 3, using their own previous rings.

Questions at the table

- **My selected boat moved because somebody else voted. Does it still score?** Yes, at the harbor it reaches after both turns.
- **My arrow is 0. Do I skip scoring or crossing?** No. Choose, score and cross normally.
- **Can both of my boats share one berth?** Yes. You still select only one. Inspect their A/B labels when needed.
- **Can five selected boats share a harbor?** Yes; they all earn zero there because no reward exceeds four. Nobody is blocked or removed.
- **Which direction remains in the third round of a tide?** The one you have not used. It is public information; your A/B choice stays secret until reveal.

- **A stack hides an identity or feels unstable.** Pause with all positions unchanged, read and record that berth's identities, and carefully separate the stack to inspect it. Restore the position before continuing. During a ring turn, support a tall stack lightly without changing its berth. If handling cannot preserve the position, stop and correct the physical issue; there is no stack-height penalty in the rules.
- **Somebody played a direction already used this tide.** Before moving rings, return all revealed pairs and choose the round again with legal cards. Information already revealed cannot be undone; this is an error recovery procedure, not a strategy. If an error is found later, restore the last agreed position from the score sheet and players' records before replaying.

First decisions to try

A quiet reward-three harbor can pay more than a crowded reward-four harbor. Watch which boat your opponents can choose, which arrows they have left, and which ring your harvest will enter next. Your unselected boat is still riding the current; that position can matter next round. On the last round of a tide, everyone knows the remaining directions but still chooses between two boats.

Adjust the session length

Agree before setup to use **6 rounds** for an introductory game, **12 rounds** for the standard game, or **18 rounds** for a longer game. Finish only at the end of a tide. Refresh directions after every third round except the final one, and use the same scoring and shared-win rule. The 6- and 18-round variants have not been included in the recorded simulation experiments; their duration and balance are untested. No fixed number of minutes is promised for any version.

What has been checked

Make verification includes 1,640 seeded, complete 12-round computer games and an audit of 19,680 recorded transitions against a separately formulated reference engine. Eighteen edge checks cover such cases as cross-ring crowd sharing, opposing arrows, simultaneous crossing and illegal card reuse. In 120 sampled public states, the search policy chose actions averaging 2.9804 expected immediate points versus 2.5419 for uniform choices; the exact best average was 2.9941 under the same uniform-opponent assumption.

These checks support executable rules and useful choices in those sampled states. They do not establish human enjoyment or overall balance. A five-player stress experiment remains a concern: four synchronized alternating-boat bots outperformed a search policy that assumed random opponents. A responder told their pattern improved to 33.975 mean points versus their 34.1125, but its 15.42% fractional win rate in 40 games remained below the 20% equal-strength reference. That warrants further evaluation of five-player synchronization. The rules remain for 2-5 players; this evidence does not justify hiding that tension or declaring a dominant strategy.

The experiments are Make evidence in `../evidence/simulation/`, with source, seeds, traces and limitations. This Spark route has no separate Playtest stage. No human play, successful physical print, physical fit, durability, age suitability or session duration is claimed.

Care and small parts

Play on a level table. Lift and move the game only after removing loose boats; the rings lift out and are not captive. Keep the small boat tokens away from children who may mouth them. Remove a chipped or cracked piece from use. Ask an adult to handle cutting cards, print cleanup and permanent-marker labeling.